

Binary Arithmetic

Giannacco

Binary Addition

$$0 + 0 = 0$$

$$0 + 1 = 1$$

$$1 + 0 = 1$$

$$1 + 1 = 0, \text{ carry } 1$$

Binary Subtraction

$$0 - 0 = 0$$

$$0 - 1 = 1, \text{ borrow } 1$$

$$1 - 0 = 1$$

$$1 - 1 = 0$$

Binary Multiplication

$$0 \times 0 = 0$$

$$0 \times 1 = 0$$

$$1 \times 0 = 0$$

$$1 \times 1 = 1$$